

Information sources and coverage

We will search MEDLINE (via PubMed), EMBASE (via Elsevier), Web of Science Core Collection, Cochrane CENTRAL, and CINAHL from inception to September 30, 2025. We will also search ClinicalTrials.gov and the WHO ICTRP for ongoing/unpublished studies and screen reference lists and citing articles of all included records.

Supplementary AI/Recursive Searching

In addition to structured database searches, we will employ recursive and AI-assisted strategies to maximize retrieval. These will include:

- Backward citation chasing: screening the reference lists of included studies.
- Forward citation chasing: using Web of Science and/or Google Scholar to identify newer articles citing included studies.
- AI/semantic search tools: using PubMed's "Similar Articles" function, Semantic Scholar, and/or OpenAlex embeddings to identify studies conceptually related to those already eligible, even if not retrieved by the structured search.

All records retrieved through these supplementary methods will be screened using the same eligibility criteria as database-derived records. The yield of this arm will be documented separately (e.g., in the PRISMA flow diagram and supplementary tables) to maintain transparency. These methods will not replace the primary structured database search but serve as a complementary sensitivity approach.

Restrictions and Filters

Searches will be limited to human studies with no date or language restrictions; non-English articles will be translated where feasible (or listed as "awaiting classification" if translation is not possible). Publication types excluded at screening: animal studies, case reports/series, editorials, narrative reviews.

Documentation

The detailed line-by-line strategies for each database (including subject headings and keywords) are provided in the protocol appendix and will be rerun prior to final analyses to capture newly published records.